

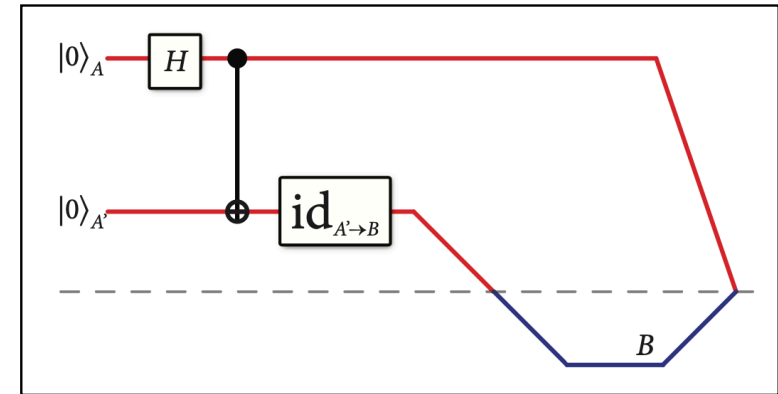


Entanglement Assisted Classical Capacity

Entanglement and Bell State Measurement



- Alice performs local operations (the Hadamard and CNOT) and consumes one use of a noiseless qubit channel to generate one noiseless ebit $|\Phi^+\rangle\langle\Phi^+|_{AB}$ shared with Bob.
- Following states form an orthonormal basis (Bell states):
 $\{ |\Phi^+\rangle_{AB}, X_A |\Phi^+\rangle_{AB}, Z_A |\Phi^+\rangle_{AB}, Z_A X_A |\Phi^+\rangle_{AB} \}$
- So these states are perfectly distinguishable: Using Bell State Measurement (BSM)
- BSM: Convert Bell States $\rightarrow$ Computational Basis
 - BSM performs controlled-NOT from A to B , Hadamard on A , and then measuring A and B in the standard basis.
 - Circuit:
 Bell state $\rightarrow$ CNOT $\rightarrow$ $H \rightarrow \{ |00\rangle, |01\rangle, |10\rangle, |11\rangle \}$
 - Transformations:
 $|\Phi^+\rangle \rightarrow 00$, $|\Phi^-\rangle \rightarrow 10$, $|\Psi^+\rangle \rightarrow 01$, $|\Psi^-\rangle \rightarrow 11$



Bell States (Orthonormal Basis for 2 Qubits):

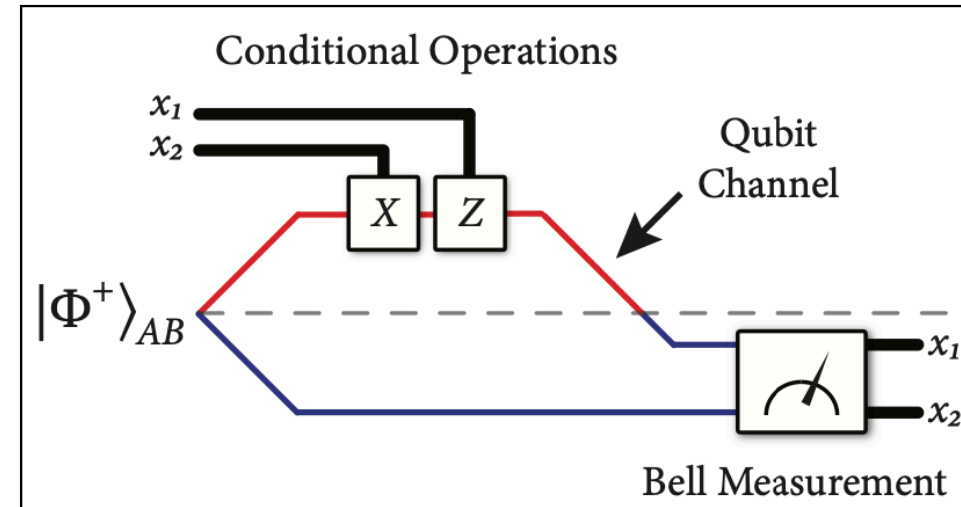
$$|\Phi^+\rangle = \frac{|00\rangle + |11\rangle}{\sqrt{2}}, \quad |\Phi^-\rangle = \frac{|00\rangle - |11\rangle}{\sqrt{2}}$$

$$|\Psi^+\rangle = \frac{|01\rangle + |10\rangle}{\sqrt{2}}, \quad |\Psi^-\rangle = \frac{|01\rangle - |10\rangle}{\sqrt{2}}$$



Superdense Coding

- Holevo Bound: Even though a qubit can exist in infinitely many states, one qubit can reveal at most one classical bit of information (without shared entanglement).
- Superdense coding doubles the classical bitrate by using pre-shared entanglement.
- Alice and Bob share an ebit: $|\Phi^+\rangle_{AB}$
- Alice would like to transmit two classical bits x_1x_2 to Bob
- Alice performs a Pauli rotation conditioned on x_1x_2 :
 $00 \rightarrow I$, $01 \rightarrow X_A$, $10 \rightarrow Z_A$, $11 \rightarrow X_AZ_A$
- Alice sends her share of the ebit over a noiseless qubit channel to Bob.
- Bob then performs a BSM to get x_1x_2
- Following states form an orthonormal basis (Bell states):
 $\{ |\Phi^+\rangle_{AB}, X_A |\Phi^+\rangle_{AB}, Z_A |\Phi^+\rangle_{AB}, Z_A X_A |\Phi^+\rangle_{AB} \}$



Bell States (Orthonormal Basis for 2 Qubits):

$$|\Phi^+\rangle = \frac{|00\rangle + |11\rangle}{\sqrt{2}}, \quad |\Phi^-\rangle = \frac{|00\rangle - |11\rangle}{\sqrt{2}}$$

$$|\Psi^+\rangle = \frac{|01\rangle + |10\rangle}{\sqrt{2}}, \quad |\Psi^-\rangle = \frac{|01\rangle - |10\rangle}{\sqrt{2}}$$



Is it Possible to Violate Holevo's Bound?

1 Qubit $\longrightarrow$ max. 1 cbit/ch. use data rate

Q: Why can't encode 2 bits in 1 Qubit

$0 \rightarrow |0\rangle$

$1 \rightarrow |1\rangle$

$\left\{ \begin{array}{ll} 00 \rightarrow |00\rangle & 10 \rightarrow |10\rangle \\ 01 \rightarrow |01\rangle & 11 \rightarrow |11\rangle \end{array} \right\}$

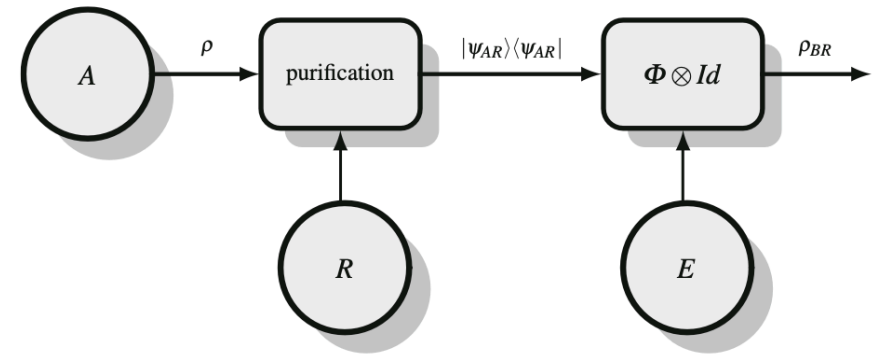
- The 4 quantum states are not mutually orthogonal, hence they can't be perfectly discriminated by Bob.
- Therefore, the two classical bits can't be sent reliably.

Quantum Mutual Information



- Purification of input state: $|\psi_{AR}\rangle$
- From theory of purification: $S(\rho_R) = S(\rho_A)$ where $\rho_R = \text{Tr}_A |\psi_{AR}\rangle\langle\psi_{AR}|$
- Output state: $\rho_{BR} = (\Phi \otimes \text{Id}_R)(|\psi_{AR}\rangle\langle\psi_{AR}|)$
- Quantum Mutual Information:

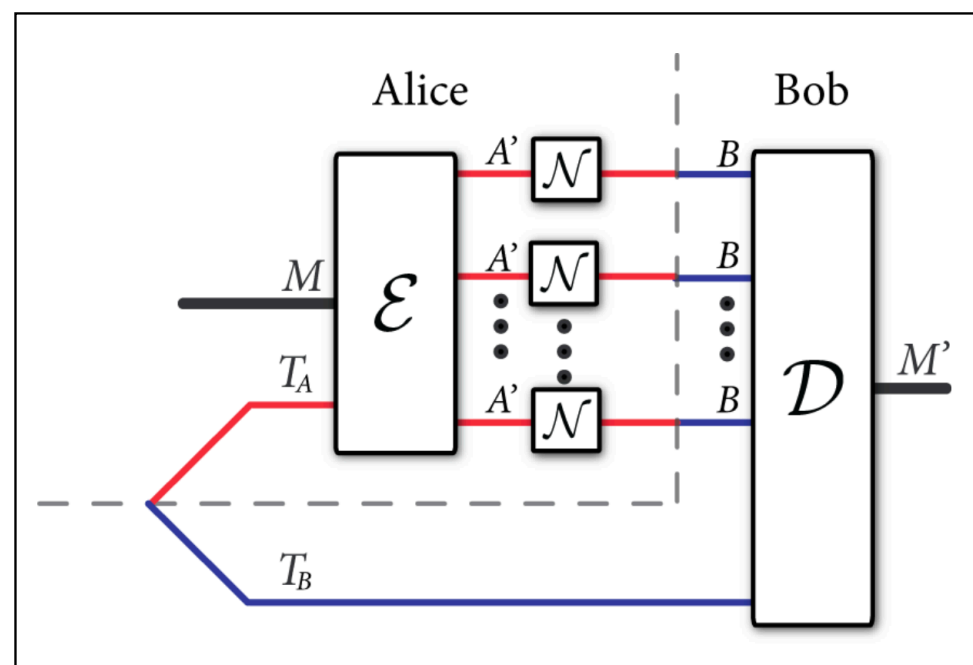
$$I(\rho, \Phi) := S(\rho_R) + S(\rho_B) - S(\rho_{BR})$$
- Depends on Input entropy: $S(\rho_R) = S(\rho_A)$, output entropy: $S(\rho_B) = S(\Phi(\rho_A))$ and joint entropy $S(\rho_{BR})$
- It can be shown that $S(\rho_{BR}) = S(\rho_E)$
- Entropy Exchange: $S(\rho, \Phi) := S(\rho_{BR}) = S(\rho_E)$
- Quantum Mutual Information: $I(\rho, \Phi) = S(\rho_A) + S(\Phi(\rho_A)) - S(\rho, \Phi)$.
- Quantum Mutual Information: Applicable for Entanglement assisted classical capacity evaluation



Entanglement Assisted Classical Capacity



- Super-dense coding: Entanglement can double the classical capacity of a noiseless qubit channel.
- Generalisation of Superdense Coding to arbitrary channels: Entanglement Assisted Classical Capacity
- Alice & Bob share copies of pure entangled state: $\Psi_{T_A T_B}$
- Entanglement Assisted Capacity (Maximum of Quantum Mutual Information) :
- $C_E(\mathcal{N}) = I(\mathcal{N}) = \max I(A; B)_\rho$ where $\varphi_{AA'}$ is a pure bipartite state and $\rho_{AB} = \mathcal{N}_{A \rightarrow B}(\varphi_{AA'})$
- A' is the system transmitted through the channel (red wire in the figure), while A is an untouched reference/purifying system used to quantify how much information the channel preserves.





Entanglement Assisted Classical Capacity Examples

- Ex-1: Identity (Noiseless) Qubit Channel: $\mathcal{N}(\rho) = \rho$
- Ex-2: Completely Depolarising Qubit Channel: $\mathcal{N}(\rho) = \frac{I}{2}$
- Ex-3: Bit Flip Channel: $\mathcal{N}(\rho) = (1 - p)\rho + pX\rho X$